



Completing the Square & the Quadratic Formula

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Learning Objectives

- Identify and construct perfect square trinomials
- Solve quadratic equations by completing the square
- Solve quadratic equations using the quadratic formula
- Solve quadratic inequalities using a sign chart

Solve each problem completely, showing all steps and writing answers in exact form.

1. Warm-up: Solve the quadratic equation by factoring.

$$x^2 - 5x + 6 = 0$$

Answer: _____

2. Determine the value of c that makes the expression a perfect square trinomial.

$$x^2 + 12x + c$$

Answer: _____

3. Determine the value of c that makes the expression a perfect square trinomial.

$$x^2 - 7x + c$$

Answer: _____

4. Factor the perfect square trinomial.

$$x^2 - 10x + 25$$

Answer: _____

5. Solve the quadratic equation by completing the square.

$$x^2 + 6x - 7 = 0$$

Answer: _____

6. Solve the quadratic equation by completing the square.

$$2x^2 - 8x + 2 = 0$$

Answer: _____

7. Solve using the quadratic formula.

$$x^2 + 5x + 3 = 0$$

Answer: _____



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8. Solve using the quadratic formula.

$$3x^2 - 2x - 1 = 0$$

Answer: _____

9. Solve the quadratic inequality using a sign chart.

$$x^2 - x - 6 > 0$$

Answer: _____

10. Solve the quadratic inequality using a sign chart.

$$x^2 + 2x - 8 \leq 0$$

Answer: _____





This worksheet covers the topics in the video: warm-up factoring/quadratic review, perfect square trinomials (PST), creating a PST by completing the square, solving quadratics by completing the square, deriving and applying the quadratic formula, and solving quadratic inequalities using a sign chart.

Solutions

1. Warm-up: Solve the quadratic equation by factoring.

$$x^2 - 5x + 6 = 0$$

- Factor the trinomial as the product of two binomials whose constants multiply to six and add to negative five.
- The factored form is the quantity x minus two times the quantity x minus three equals zero.
- Set each factor equal to zero.
- The solutions are x equals two and x equals three.

Answer: $x = 2, x = 3$

2. Determine the value of c that makes the expression a perfect square trinomial.

$$x^2 + 12x + c$$

- Take half of the coefficient of x , which is six.
- Square six to obtain thirty-six.
- So c equals thirty-six, giving the perfect square the quantity x plus six squared.

Answer: $c = 36$

3. Determine the value of c that makes the expression a perfect square trinomial.

$$x^2 - 7x + c$$

- Take half of negative seven to get negative seven-halves.
- Square negative seven-halves to obtain forty-nine fourths.
- So c equals forty-nine over four.

Answer: $c = \frac{49}{4}$

4. Factor the perfect square trinomial.

$$x^2 - 10x + 25$$

- Identify that twenty-five is the square of five and that ten is twice five.
- Write the trinomial as the quantity x minus five squared.

Answer: $(x - 5)^2$



5. Solve the quadratic equation by completing the square.

$$x^2 + 6x - 7 = 0$$

- Move the constant to the right side to get x squared plus six x equals seven.
- Take half of six to get three, then square it to get nine.
- Add nine to both sides, giving x squared plus six x plus nine equals sixteen.
- Factor the left side as the quantity x plus three squared equals sixteen.
- Take the square root of both sides to get x plus three equals plus or minus four.
- Subtract three from both sides to get x equals one or x equals negative seven.

Answer: $x = 1, x = -7$

6. Solve the quadratic equation by completing the square.

$$2x^2 - 8x + 2 = 0$$

- Divide every term by two to get x squared minus four x plus one equals zero.
- Move the constant to the right side to get x squared minus four x equals negative one.
- Take half of negative four to get negative two, then square it to get four.
- Add four to both sides, giving x squared minus four x plus four equals three.
- Factor the left side as the quantity x minus two squared equals three.
- Take the square root of both sides to get x minus two equals plus or minus the square root of three.
- Add two to both sides to get x equals two plus or minus the square root of three.

Answer: $x = 2 \pm \sqrt{3}$

7. Solve using the quadratic formula.

$$x^2 + 5x + 3 = 0$$

- Identify a equals one, b equals five, and c equals three.
- Substitute into the quadratic formula.
- The discriminant is twenty-five minus twelve, which equals thirteen.
- The solutions are x equals negative five plus or minus the square root of thirteen, all over two.

Answer: $x = \frac{-5 \pm \sqrt{13}}{2}$

8. Solve using the quadratic formula.

$$3x^2 - 2x - 1 = 0$$

- Identify a equals three, b equals negative two, and c equals negative one.
- Substitute into the quadratic formula.
- The discriminant is four plus twelve, which equals sixteen, and its square root is four.
- The solutions are x equals two plus four over six, which simplifies to one, and x equals two minus four over six, which simplifies to negative one third.

Answer: $x = 1, x = -\frac{1}{3}$



9. Solve the quadratic inequality using a sign chart.

$$x^2 - x - 6 > 0$$

- Factor the quadratic as the quantity x minus three times the quantity x plus two.
- Find the critical numbers where the expression equals zero, which are x equals three and x equals negative two.
- Use a sign chart to test the intervals to the left of negative two, between negative two and three, and to the right of three.
- The expression is positive on the outer intervals, so the solution is x less than negative two or x greater than three.

Answer: $x < -2$ or $x > 3$

10. Solve the quadratic inequality using a sign chart.

$$x^2 + 2x - 8 \leq 0$$

- Factor the quadratic as the quantity x plus four times the quantity x minus two.
- Find the critical numbers where the expression equals zero, which are x equals negative four and x equals two.
- Use a sign chart to test the intervals; the expression is negative between the critical numbers.
- Include the endpoints since the inequality is less than or equal to, giving negative four less than or equal to x less than or equal to two.

Answer: $-4 \leq x \leq 2$

